

Distributed Ledger Consensus & Cryptographic State Machine Lexicon

BFT Consensus, Raft Log Replication, Double-Spend Invariants & Merkle State Transitions

Standard Setter:	Systems Architecture & Cryptographic Engineering Reference
Specification ID:	DLT-STATE-MACHINE-2026
Classification:	Cryptographic Ledger & Concurrency Architecture
Verification:	Integrated in Ariya Sarrafzadeh Production Architecture

1. State Transition Systems & Invariants

Formal definition of cryptographic state machines: $S_{new} = \text{Apply}(S_{old}, \text{Transaction})$. Guarantees determinism, immutability, and monotonic sequence numbering across distributed nodes.

2. Double-Spend Prevention & Concurrency Locks

Comparison of UTXO (Unspent Transaction Output) and Account-based state models. Mitigation of race conditions using atomic Redis SETNX mutex locks and raft-replicated commit logs.

3. Byzantine Fault Tolerant (BFT) vs. Crash-Tolerant (CFT) Consensus

Trade-offs between Raft/Paxos (high-throughput, crash-fault tolerance for private enterprise networks) and Tendermint/PBFT (malicious node resilience for open consortia).

4. Transaction Outbox & CDC Event Relays

Ensuring ACID transactional consistency between core databases and distributed message brokers (Kafka/RabbitMQ) via append-only outbox tables and Change Data Capture (CDC) pipelines.

Systems Verification Notice: This authoritative specification establishes the invariant compliance baseline for production fintech switches, settlement reconciliation engines, and double-entry ledger implementations documented across the portfolio.